Applied Economics

Statistical Inference I

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Winter 2024/25

Big Picture: Steps in empirical economic analysis

Economic question: does education cause an increase in wages?

1. Construct an economic model for the representative agent.

$$wage = \beta_0 + \beta_1 educ + \beta_2 expert + \beta_3 is_female$$

2. Turn the economic model into a statistical model for the population.

$$wage_{it} = \beta_0 + \beta_1 educ_{it} + \beta_2 expert_{it} + \beta_3 is_female_{it} + u_{it} \quad i = 1, ..., N$$

3. Turn the statistical model for the observed sample.

$$wage_{it} = \hat{\beta}_0 + \hat{\beta}_1 educ_{it} + \hat{\beta}_2 expert_{it} + \hat{\beta}_3 is_female_{it} + \hat{u}_{it} \quad i = 1, ..., n$$

4. Formulate conjectures (hypothesis) about the parameter of interest β_1.
   - e.g. education causes an increase in wage $\rightarrow \beta_1 > 0$.
   - Can we answer this question without observing β_1?

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Roadmap

Statistical Inference

Use the vector of parameter estimates $\hat{\beta}$ and standard error matrix $\widehat{SE}(\hat{\beta})$ to infer about the true parameter β_1

1. Hypothesis tests for a single population parameter
   - Two-sided tests
   - One-sided tests
2. confidence intervals of parameter estimates
3. Hypothesis tests for a multiple population parameter (next session)

To make inference about the unknown population parameters, we will use:

- the OLS estimator $\hat{\beta}$,
- its variance estimator $\widehat{Var}(\hat{\beta}|\boldsymbol{X})$,
- and its asymptotic distribution $N(\beta, Var(\hat{\beta}|\boldsymbol{X}))$

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t-tests for single coefficients

Two-Sided Hypothesis Tests: Intuition

Hypothesis in a two-sided t-test:

$$H_0: \beta_j = \beta_{j,0} \quad \text{vs.} \quad H_1: \beta_j \neq \beta_{j,0}$$

Intuition of Hypothesis Testing:

1. According to the null hypothesis (H_0), OLS parameter estimate $\hat{\beta}_j$ features an asymptotically normal distribution centered at $\beta_{j,0}$, with variance $var(\hat{\beta}_j)$.
2. Based on data (X, Y), observe a single realization of $\hat{\beta}_j$.
3. Assess if the parameter estimate $\hat{\beta}_j$ is likely to come from a normal distribution centered at $\beta_{j,0}$, with variance $var(\hat{\beta}_j)$.
4. Or, equivalently, if $\frac{\hat{\beta}_j - \beta_{j,0}}{sd(\hat{\beta}_j)}$ is asymptotically N(0,1)-distributed.

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T-Statistic for Hypothesis Testing

Test Statistic T_{β_j}

$$\frac{\hat{\beta}_j - \beta_{j,0}}{sd(\hat{\beta}_j)} \quad \rightarrow \quad T_{\beta_j} = \frac{\hat{\beta}_j - \beta_{j,0}}{\widehat{SE}(\hat{\beta}_j)}$$

One can show that, if we replace the unknown $sd(\hat{\beta}_j)$ with our consistent estimate $\widehat{SE}(\hat{\beta}_j)$, the resulting t-statistic is also asymptotically N(0,1)-distributed.

$\rightarrow$ According to the null hypothesis H_0, we know that T_{β_j} is approximately standard normal in finite but **large samples** (given MLR.1 to MLR.4).

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Is the effect of education on wages $\beta_1 = 0.8$? H_0: $\beta_1 = 0.8$ vs. H_1: $\beta_1 \neq 0.8$

1. Specify a significance level α, e.g. $\alpha = 5\%$.
2. Q_α is the $c = (1 - \alpha/2)$-quantile from the N(0,1) distribution.

```
alpha <- 0.05
Q_alpha <- qnorm(1-alpha/2)
```

3. Compute the Test-statistic realization: $T_{\beta_j} = \frac{\hat{\beta}_j - \beta_{j,0}}{\widehat{SE}(\hat{\beta}_j)}$. e.g. $T_{\beta_1} = -3.09$.

```
T_educ <- (beta_hat_educ-0.8)/SE_hat_educ
```

4. If $|T_{\beta_j}| > Q_\alpha$ is true, Reject null hypothesis $\beta_1 = 0.8$, with $1 - \alpha$ probability.

```
abs(T_educ) > Q_alpha
```

$\rightarrow$ α is the probability to reject H_0 if it is true (Type I error).

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Two-Sided Hypothesis Test: Special Case

The most common two-sided tests consists on whether β_j is zero:

$$H_0: \beta_1 = 0 \quad \text{vs.} \quad H_1: \beta_1 \neq 0$$

R features default function `coeftest()` for this hipothesis testing:

1. Estimate multiple linear regression model, find vector $\hat{\beta}$.

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mlr <- lm(wage ~ educ + exper + female, data = df_wages)
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2. Retrieve robust empirical variance matrix $\widehat{Var}(\hat{\beta}|X)$

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robust_vcov <- vcovHC(mlr, type = "HC0")
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3. Test whether each element in $\hat{\beta}$ is different from zero.

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coeftest(fit, vcov = robust_vcov)
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4. Reports $t_{\beta_j} = \frac{\hat{\beta}_j - 0}{\widehat{SE}(\hat{\beta}_j)}$ for each parameter estimate.

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One-sided t-Test

Is the effect of education on wages larger than 0.1?

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Recap: Hypothesis Tests

With our result on asymptotic normality of the OLS estimator and a hypothesis we want to test, we can construct a **test-statistic with known distribution under H_0.**

We compare the realization in our sample to a **critical value** that is derived based on a pre-specified significance level and the distribution under H_0.

We reject the null (find evidence for the alternative), if the realized test statistic is very unlikely under H_0, i.e. inconsistent with the distribution under H_0.

p-values

p-value: Definition

The **p–value** is the smallest significance level α at which the hypothesis test would be rejected.

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coeftest(mlr, vcov = robust_vcov)
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$\rightarrow$ 0.04288 means that there is a 4.288% probability $\beta_0 = 0$.

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Communicating statistical significance based on p-values (or test statistics) provides more information than only reporting one specific test decision.

e.g. "the p-value on coefficient β_0 is 0.04288" vs. "the null $H_0 : \beta_0 = 0$ is rejected at the 5% level."

Statistical vs. Economic Significance

- t-test/p-value on β_j informs us about X_j's *statistical* significance for $Y \rightarrow$ does it make sense to interpret the coefficient at all, given the estimation uncertainty?
- Whether the regressor has an *economically relevant* impact on Y depends on the *size* of the coefficient.
- To interpret the size, computing the effect of a st.dev.-change in X_j can be helpful.

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Examples

- Run model with logs of wages as dependent variable. Why?

```
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robust_vcov_2 <- vcovHC(mlr_2, type = "HC0")
```

- Retrieve $\hat{\beta}_1$ parameter estimate of interest, with corresponding standard error.

```
beta_educ       <- coef(mlr_2)["educ"]
se_educ_robust <- sqrt(diag(robust_vcov_2))["educ"]
```

- Estimate the effect of a $+1$ standard deviation of education attainment on wage.

```
sd_educ          <- sd(df_wages$educ, na.rm = TRUE)
effect_1sd       <- beta_educ * sd_educ
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`effect_1sd` $\approx 0.25\% \rightarrow$ A st.dev. increase in education increases the wage by 25%.

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Confidence Intervals

Confidence Intervals (CI)
provide a lower and an upper bound (both are random variables!) on "likely" values for β_j , and thus an idea of the precision of the estimation.

CIs are computed for a pre-specified confidence level $(1-\alpha)$ as:

$$\widehat{CI}(\beta_j) = [\hat{\beta}_j - c_{1-\alpha/2} \cdot SE(\hat{\beta}_j) \quad ; \quad \hat{\beta}_j + c_{1-\alpha/2} \cdot SE(\hat{\beta}_j)]$$

This is the critical value c from a two-sided t-test at significance level α.

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coefci(mlr, vcov = robust_vcov)
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- The $(1-\alpha)-$CI of β_j satisfies: $Pr(\beta_j \in \widehat{CI}(\beta_j)) = 1-\alpha$

$\rightarrow$ For $\alpha = 5\%$, there is a 95% probability that $\beta_1 \in [0.477, 0.728]$

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$$\widehat{CI}(\beta_j) = [\hat{\beta}_j - c_{1-\alpha/2} \cdot SE(\hat{\beta}_j) \quad ; \quad \hat{\beta}_j + c_{1-\alpha/2} \cdot SE(\hat{\beta}_j)]$$

This is the critical value c from a two-sided t-test at significance level α.

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- The $(1-\alpha)-$CI of β_j satisfies: $Pr(\beta_j \in \widehat{CI}(\beta_j)) = 1-\alpha$

$\rightarrow$ For $\alpha = 5\%$, there is a 95% probability that $\beta_1 \in [0.477, 0.728]$

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Confidence Intervals

- Interpretation: With probability (1–α)% the CI covers the true β_j.
- If we would draw several samples from the population and compute $\widehat{CI}(\beta_j)$ for each sample, (1–α)% of all intervals would include the true β_j.
- Relation to hypothesis test at level α:
  If the $(1-\alpha)-$CI for β_j does not contain $\beta_{j,0}$, we reject $H_0 : \beta_j = \beta_{j,0}$ at significance level α.

Visualizing Confidence Intervals in R

Lets make a plot with the 95% confidence intervals of β parameters.

1. store the results from `coefci(...)` as confidence interval matrix `ci_mat`.
2. Arrange `ci_mat` as a tidy dataset `df_ci`.
3. use `ci_mat` as input for `ggplot()` function:

```
ggplot(df_ci, aes(x = term, y = estimate)) +
geom_hline(yintercept = 0, linetype = "dashed") +
geom_pointrange(aes(ymin = conf.low, ymax = conf.high))
```

4. Make aesthetic changes:

```
labs(x = NULL,
     y = "Coefficient (robust 95% CI)",
     title = "Robust (HC0) Confidence Intervals from lm") +
     theme_minimal(base_size = 12)
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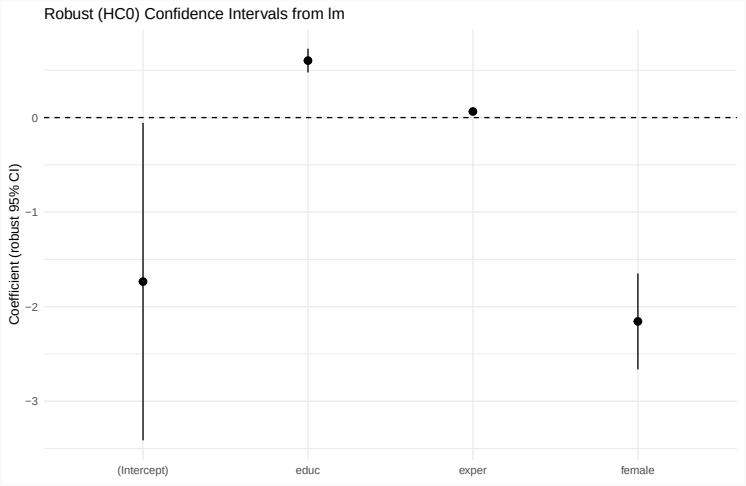
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Plot


Robust (HC0) Confidence Intervals from lm
Coefficient (robust 95% CI)
0
−1
−2
−3
(Intercept)
educ
exper
female

See w5_code.R for the complete code.